

Are FM MAS modelled on human teams?

❓ Question

- Do foundation-model multi-agent systems borrow their structure from human teams, and how far does the analogy hold?

📄 Answer

- Partly, and only for one cluster of the corpus. A specific family of frameworks deliberately transplants a human team or organization; the rest is organized by abstract communication structure or built as a research testbed, with no human-team referent.
- The clearest cases compile a known human workflow into agent roles, a fixed order, and standardized handoffs — a [Standardized Operating Procedure](#). [MetaGPT](#) runs a simulated software company (Product Manager → Architect → Project Manager → Engineer → QA Engineer), each role emitting a standardized artefact for the next ([Hong2024MetaGPTMP](#); [sec. 3.1, p. 4](#)); [ChatDev](#) mirrors a software-development waterfall ([Qian2024ChatDevCA](#); [sec. 3.1, p. 15176](#)); [MedAgents](#) runs a simulated multidisciplinary clinical consultation ([Tang2024MedAgentsLL](#); [sec. 2, p. 601](#)).
- This is not a one-off: dividing work by domain role is one of the three collaboration strategies [Tran2025](#) identifies, and the canonical illustration it gives is exactly a software team — product manager, architect, developer, QA ([Tran2025MultiAgentCM](#); [sec. 4.3.2, p. 14](#); [fig. 4, p. 15](#)).
- The motivation is reliability: adopting a workflow humans already refined, to reduce unproductive collaboration ([Hong2024MetaGPTMP](#); [sec. 1, p. 3](#)).
- But the analogy is shallow at a load-bearing point: in most of these "teams" the agents are one frozen model wearing different role prompts, so they share weights rather than being genuinely independent teammates. [MedAgents](#)'s own error analysis finds a major failure is the model simply lacking the needed domain knowledge, which the multi-expert consultation does not repair ([Tang2024MedAgentsLL](#); [sec. 4.6, p. 605](#)) — a [correlated-errors](#) pattern: same-backbone personas can agree because they share a knowledge gap, not because they are right.

📄 Scope

- Covers the frozen-LLM FM-MAS corpus captured in the wiki. The human-team model applies to the role-specialized framework cluster (software-company and clinical-consultation designs), not to debate frameworks, social-deduction testbeds, or the abstract topology taxonomies.
- The [Society of Mind](#) lineage ([Wu2024AutoGenEN](#) cites it to motivate [nested chat](#); [sec. 2, p. 6](#)) is a related-sounding but distinct framing: it is a cognitive architecture about parts inside one mind, not a model of how whole agents form a team.
- Out of scope: human-machine teaming (FM as a human's teammate), which does import human-team factors but is a different sense of the question.

✓ Evidence

- Human-organizational structure: [Hong2024MetaGPTMP](#) ([sec. 3.1, p. 4](#)) and [Qian2024ChatDevCA](#) ([sec. 3.1, p. 15176](#)) encode software-company workflows; [Tang2024MedAgentsLL](#) ([sec. 2, p. 601](#)) encodes a clinical consultation.
- Role division by domain specialty, illustrated by a software team: [Tran2025MultiAgentCM](#) ([sec. 4.3.2, p. 14](#); [fig. 4, p. 15](#)).
- The shallow-analogy caution (the consultation cannot repair knowledge the model lacks): [Tang2024MedAgentsLL](#) ([sec. 4.6, p. 605](#)).

⚠ Tensions

- The reliability payoff of borrowing human structure is unproven in the corpus: no framework isolates the role split from the extra compute it spends, and [Cemri2025WhyDM](#) finds MAS failures rooted in system design, with even targeted interventions buying bounded gains ([sec. 5.3, p. 9](#)).
- The human-team label is a designer's metaphor, not a demonstrated mechanism: a software-company role split may help because of task decomposition, not because it mirrors a human org, and no source separates the two.

🔄 What Would Change This Answer

- An ablation isolating the human-org role structure from task decomposition and added compute, showing the human-team framing itself carries the benefit.
- A framework that builds genuinely independent agents (different backbones or fine-tuned checkpoints), removing the [correlated-error](#) objection and making the team

analogy substantive rather than superficial.

🔍 Open Questions

- Role-specialized designs recur across the framework corpus — does that reflect that human-team structure works, or only that it is the easiest structure to specify with [role prompting](#)?
- Which human workflows transfer? The waterfall [SOP](#) has no path back to an earlier stage, so it may suit only tasks whose structure is already waterfall-shaped.

🔗 Connections

- The mechanism behind the human-role designs is [role-based strategy](#), realized through [role prompting](#) on a shared backbone.
- "Transferring human-organizational structure for reliability" is the first of four motivations in [why build foundation-model multi-agent systems?](#); this page develops that motivation into a standalone yes-and-no answer.
- The shallow-analogy caution connects to [does multi-agent structure actually help?](#), where [correlated errors](#) and matched-compute parity undercut the multi-agent premise.

||| Sources

- [Hong2024MetaGPTMP](#)
- [Qian2024ChatDevCA](#)
- [Tang2024MedAgentsLL](#)
- [Tran2025MultiAgentCM](#)
- [Cemri2025WhyDM](#)
- [Wu2024AutoGenEN](#)